

Alpha Main Peripheral Question Analysis.

Q. What is communication model in IoT? Describe different communication model with diagram?

Answer:

The communication model in IoT defines how devices exchange data with each other and external systems.

There are four main communications model used in IoT:-

1. Client-Server Model: This is the most common model, similar to how web browsing works. Here, a client device (sensor, actuator) sends request to a client server for information or instructions. The server process the request and sends back a response.

See and Practice diagram
from slide

Example: Sending a spreadsheet to the printer.

2. Publisher-Subscriber Model: This model involves three actors:

- ▢ Publishers (data providers)

- ▢ Subscribers (data consumers)

- ▢ broker (message router)

Publishers broadcast data to the broker, which then distributes it to ~~the~~ interested subscribers.

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Example: Sensor networks where multiple devices (subscribers) receive data updates from a single source (publisher) like a temperature sensor

3. Push-Pull Model: This model combines aspects of client-server and publish-subscribe. Devices can actively request data (Pull) or receive updates

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sent from another device (Push) through a central message queue.

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Example: Home automation system where sensors push data updates and receive control commands from a central hub.

7. Exclusive Point Model: This model establishes a direct, two-way communication channel between two devices for continuous data exchange.

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Example: Industrial control systems where machines communicate directly with each other to coordinate tasks.